

# SoilLoop-2: Part Alpha

Core formulation and scenarios S0-S2

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## Purpose

SoilLoop-2 represents soil organic carbon as a fast pool and a protected pool. The note compares turnover time under three moisture response assumptions without treating any scenario as a forecast.

$$dC_f/dt = I - k_f f(\theta) C_f$$

$$dC_p/dt = a k_f C_f - k_p C_p$$

## Scenario definitions

S0 fixes moisture response at unity. S1 applies a piecewise linear response. S2 adds a saturation plateau but retains the same carbon inputs.

## Parameter table

Parameter values are synthetic and retain units explicitly so that cross-document use can be audited.

| Parameter | Meaning            | Value | Unit                           |
|-----------|--------------------|-------|--------------------------------|
| k_f       | fast turnover      | 0.084 | day <sup>-1</sup>              |
| k_p       | protected turnover | 0.006 | day <sup>-1</sup>              |
| a         | transfer fraction  | 0.23  | 1                              |
| θ_ref     | reference moisture | 0.41  | m <sup>3</sup> m <sup>-3</sup> |

## Moisture response

The moisture response changes the effective loss rate while leaving the named pools intact. Reported turnover time is conditional on the selected response curve.

Footnote A. The value for θ\_ref is an internal scenario constant, not a field estimate.

## Alpha conclusions

S2 delays loss from the fast pool during the saturation plateau, whereas S1 responds immediately to decreasing moisture. Neither scenario changes the protected-pool definition.

## Data record

The complete synthetic parameter set is identified by accession SLC-26-09. Part Beta uses the same definitions and must be read without terminology drift.

## References

- [1] Park J. Conditional turnover time in two-pool models. *Soil Systems* 14:61-70.
- [2] Adebayo K. Moisture response functions without forecast claims. *Model Notes* 5:9-16.